By TOUCH DEN · November 11, 2026

What a Computer Does

Video 1 · What a Computer Is, and Its Four Jobs

How Computers Work · Section 1

In this video, you will learn:

- What a computer really is, under the surface
- The four things every computer does
- Hardware versus software

What a computer is

A computer is a machine that follows a list of instructions, step by step, very fast, to work with information.

- It does not think or understand. It follows instructions, literally, billions of times a second.
- Everything it seems to ”do” is a vast number of tiny, simple steps, done so fast it looks instant.

Two powers: **speed** (billions of steps a second) and **reliability** (exactly what the instructions say).

This is also the security problem: a computer follows *whatever* instructions it is given, **including an attacker’s**.

The four jobs

Every computer, from a phone to a server, does four things:

Stage	Means	Example
Input	Information comes in	Typing, a network message, a sensor
Processing	It works on it, following instructions	Calculating, deciding
Output	A result goes out	The screen, a reply
Storage	Kept for later	Saving a file

When you meet an attack later, ask: **which of the four is it attacking?**

Hardware and software

- **Hardware:** the physical machine, the parts you can touch. Processor, memory, disk, screen.
- **Software:** the instructions, the programs that tell the hardware what to do. An OS, an app, a virus.

Software is a list of instructions; hardware is the machine that carries them out.

Three layers: **hardware** at the bottom, the **operating system** managing it (lesson 5), your **apps** on top. Attacks hit every layer, especially the OS in the middle, whoever controls it controls everything above.

Recap: what a computer is

- A computer follows numeric instructions, step by step, very fast, it doesn't "understand".
- The four jobs: **input, processing, output, storage**.
- **Hardware** is the machine; **software** is the instructions; the **OS** sits between.

Next video: why everything inside is numbers, and why that matters for security.

What a Computer Does

Video 2 · Numbers, Instructions, and Why It Matters

How Computers Work · Section 1

In this video, you will learn:

- Why everything inside a computer is numbers
- Why instructions and data are stored the same way
- Why security people must understand the machine

Everything inside is numbers

Inside a computer, **everything is numbers**, not just sums.

- A letter of text is a number (lesson 3).
- A color in a photo is numbers. A sound is a list of numbers.
- A program's instructions are numbers (lesson 8).

The computer has no idea what a number "means", the same number could be a letter, a brightness, or an instruction.

And the numbers are stored as huge quantities of **on-or-off switches**, patterns of 1s and 0s (next lesson).

Instructions and data are the same stuff

A computer stores two different things the same way:

- **Data:** the information a program works on (a photo, a message).
- **Instructions:** the steps the program follows (the program itself).

Both are just numbers in the same memory, the **stored-program** idea.

- **Good:** to do something new, you just give it a new program, not a new machine.
- **Dangerous:** a computer can be tricked into treating **data as instructions**, the root of buffer overflows and code injection (later courses).

Why security needs the machine

You cannot secure, attack, or investigate what you do not understand.

Security work	Machine knowledge it needs
Reverse engineering	Reading a program's instructions
Binary exploitation	Bending memory, especially the stack
Malware analysis	Executables, processes, the OS
Digital forensics	Recovering data from memory and disks

This course is the trunk; those courses are the branches.

Recap: numbers and why it matters

- Inside a computer, **everything is numbers**, stored as on/off switches.
- Instructions and data share the same memory (**stored-program**): flexible, but attackable.
- Every later security course, reversing, exploitation, forensics, rests on this machine knowledge.

Next lesson: the ons and offs themselves, bits, binary, and hexadecimal.